

From Biosensors to Programmable Biological Decision Units

As early as 1961¹, studies demonstrated that logic gates and other computational functions could be implemented using genetically encoded circuits. These gave rise to biosensors—specifically whole-cell biosensors (WCBs)—a class of synthetic organisms which can detect the presence of a specified target, and then release a detectable signal². A classic example is arsenic WCBs, which can detect toxic levels of arsenic and fluoresce³. While WCBs are useful, they have limited autonomy due to their reliance on external agents to interpret and act on their signals.

Microbial Edge Computing (MEC) is the logical extension: by engineering WCBs to respond directly to detected signals, they become programmable biological decision units (PBDUs). An arsenic PBDU would build off of the WCB, but instead of just releasing a reporter when arsenic is detected, it would go a step further and trigger a bioremediation response. This reflects edge computing principles, where data is processed locally rather than offloaded to a central server first. For example, a camera that detects motion locally and only uploads relevant footage to the cloud, instead of uploading everything to be processed.

Recent advances in synthetic organisms, advanced gene circuits, and microbial communication make creating PBDUs feasible with modern technology. By consolidating these tools into a standardized framework, MEC could introduce a new class of autonomous, field-deployable, multi-functional biological devices with applications ranging from environmental sensing systems to real-time gut diagnostics.

Embedding Logic in Living Systems

For microbe PBDUs to function as actuators in MEC systems, they require genetic circuits to interpret information locally, a mechanism to store information, and molecular communication systems to relay information to other organisms or devices. A PBDU must be able to perform basic logical operations (AND, OR, NOT) and signal processing (attenuation, amplification, noise filtering) in order to interpret environmental stimuli to trigger a certain genetic response. Advances like CRISPRi and CRISPRa have improved the specificity, modularity, and latency of genetic circuits⁴. Temporal logic can also be included in circuits by using bacteriophage integrases⁵ and oscillatory networks⁶. For more complex MEC systems, systems like DNA Mutational Overwriting Storage (DMOS)⁷ allow PBDUs to use past experiences in their decision making process. Finally, MEC devices must be able to communicate with other biological and digital systems. PBDUs can communicate with each other using chemical signals or through processes like quorum sensing. To interface with digital systems, PBDUs can use ion-gated channels to produce and detect electrical signals, fluorescence and optogenetic signalling to send and receive optic information, as well as gas production and detection. PBDUs with communication capabilities enable MEC systems to function as an interconnected network of

biosensors and processors. All of these core functions already exist in isolation, but consolidated under MEC they can be engineered into programmable organisms capable of distributed, autonomous decision making.

Use Cases of Microbial Edge Computing

With PBDUs functioning as decentralized sensors, processors, and actuators, MEC provides a foundational architecture for both the biological Internet of Things (bio-IoT) and the biological Internet of Everything (bio-IoE). Bio-IoT refers to networks of interconnected biological and electronic devices—such as biosensors and PBDUs—that enable real-time monitoring and localized responses. Bio-IoE extends this framework by integrating not only devices, but also humans, data flows, processes, and environmental systems, transforming isolated sensing networks into adaptive, system-level infrastructure.

While biosensors have been used in the gut to detect inflammation and intestinal disease⁸, MEC enhances this capability by embedding logic and enabling autonomous actuation. In one proposed design, PBDUs are enclosed in an ingestible capsule with a semipermeable membrane, allowing them to monitor local pH, pressure, and inflammatory signals. Upon detecting a clinically relevant imbalance, the microbes would initiate a targeted therapeutic response and trigger a glucose-powered biofuel cell (BFC)⁹ to activate a wireless transmitter, sending diagnostic data or therapeutic instructions to a wearable device. Though this represents a patient-level bio-IoT application, the same architecture could scale into distributed MEC systems for environmental or infrastructural monitoring.

MEC systems offer a powerful framework for real-time, autonomous agricultural infrastructure. In one design, engineered soil PBDUs function as local processors—detecting nutrient imbalances, temperature shifts, or early signatures of pathogens. Unlike traditional biosensors, MEC-enabled microbes can compute on multi-signal conditions, coordinating responses through quorum sensing to release growth-promoting compounds only when specific stress combinations are met. These systems can also act as early warning platforms, secreting immune-priming signals to nearby crops. Output signals—such as fluorescent markers, volatile gases, or wireless transmission via integrated microelectronics—can trigger alerts or external interventions. By distributing sensing, decision-making, and actuation across a microbial network, MEC enables decentralized, scalable crop management while contributing to a broader bio-IoE ecosystem for agricultural automation and data integration.

These theoretical examples illustrate how MEC systems could elevate microbes from isolated biosensors to collaborative agents capable of distributed decision-making. By bridging microscopic sensing with macroscale networks, MEC-enabled bio-IoT and bio-IoE architectures enable programmable biological infrastructure—scalable from personal medical devices to environmental systems that span entire ecosystems.

Towards Standardized Architectures

For MEC to scale from isolated experiments into field deployable systems, standardization and modularity should be prioritized when designing components and architectures. Similar to how electrical engineers follow certain visual conventions to prototype devices, MEC system designers will need the same. This also has the added benefit of simplifying collaborative efforts and reducing the likelihood of mistakes.

A MEC schematic would be composed of three layers, each representing a different layer of abstraction. The first layer is the physical environment, which would have icons or drawings to represent PBDUs and other components. This does not have to be an exact 1:1 of the environment, it can generalize repeating systems as long as there's enough detail to fully describe the device. The next layer would be the logical layer. This layer doesn't represent a physical map, it just contains the schematics of different devices (electrical circuits, PBDU genetic circuits, computational logic, etc.), as well as any additional notes that are relevant to the conceptual design of the MEC architecture. Finally, the technical layer follows the same structure as the previous one and is used to provide exact specifications on how to build certain components and devices. For a PBDU, this layer would mention the specific details (like gene names) to building a genetic circuit. This layer would also include information like wire lengths, device specifications, or any missing details required to fully construct the circuit

Here is an outline of the architecture:

1. Physical Environment
 - a. Contains icons and the general physical setup of an MEC system
 - b. Not necessarily accurate scale
2. Logical Layer
 - a. Collection of schematics for all the logical components
 - b. Still abstract, but provides more detail for component interactions
3. Technical Layer
 - a. Technical specifications for fully constructing an MEC system

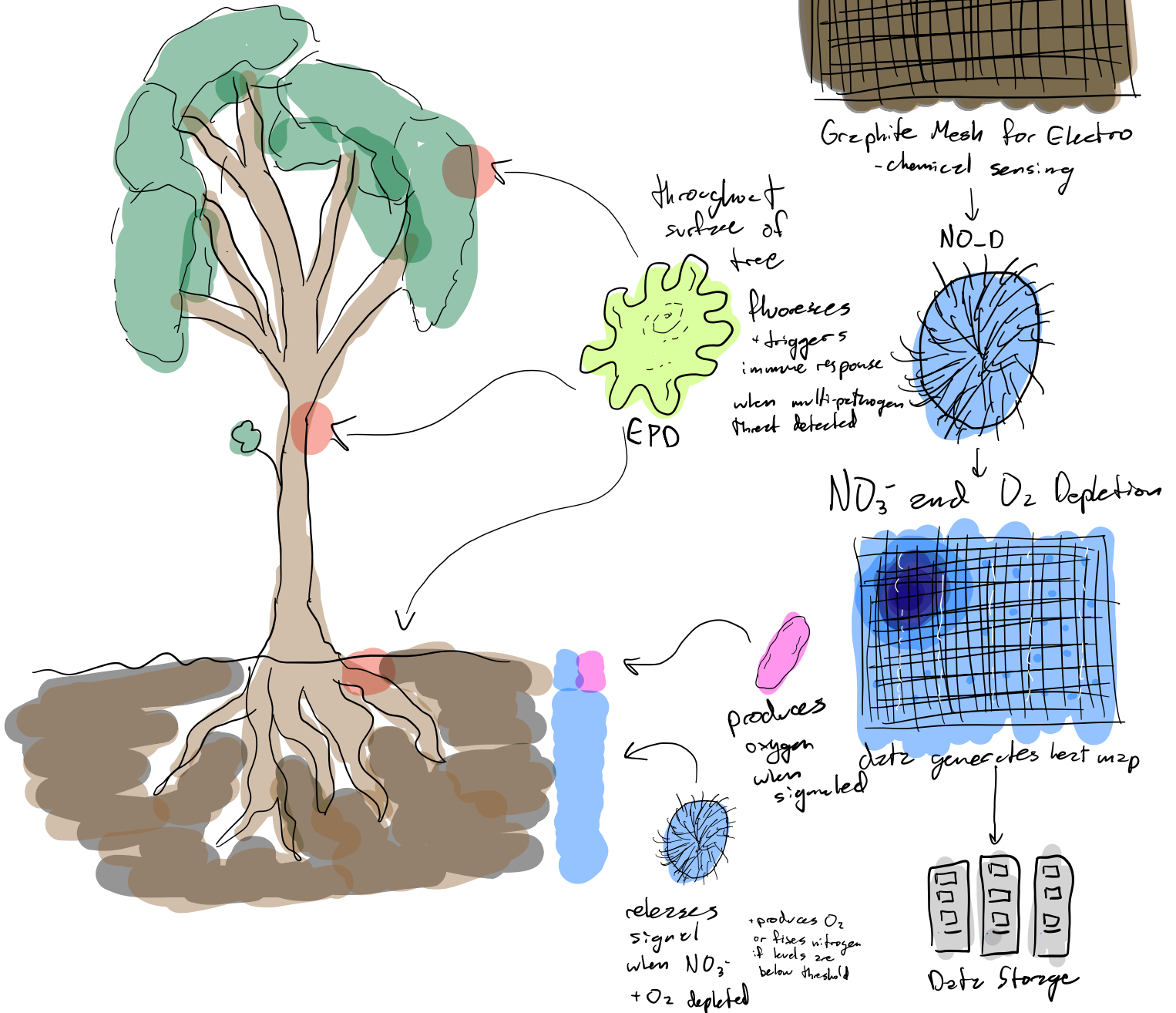
The idea behind a multi-layer architecture is to make it easier to go through the ideation and prototyping process. A conceptual MEC system can be explained with only the first layer, but as the idea develops, more layers can be included. To illustrate this, I made a simple diagram of a bio-IoE MEC system for agriculture. I abstracted away the third layer, as this is just supposed to be a general proof of concept. This framework is not specific to MEC systems—or even bio-IoE systems—but if the field develops more, so will the specificity of the diagrams.

ORCHARD BIO-IOE

MEC MONITORING

SYSTEM

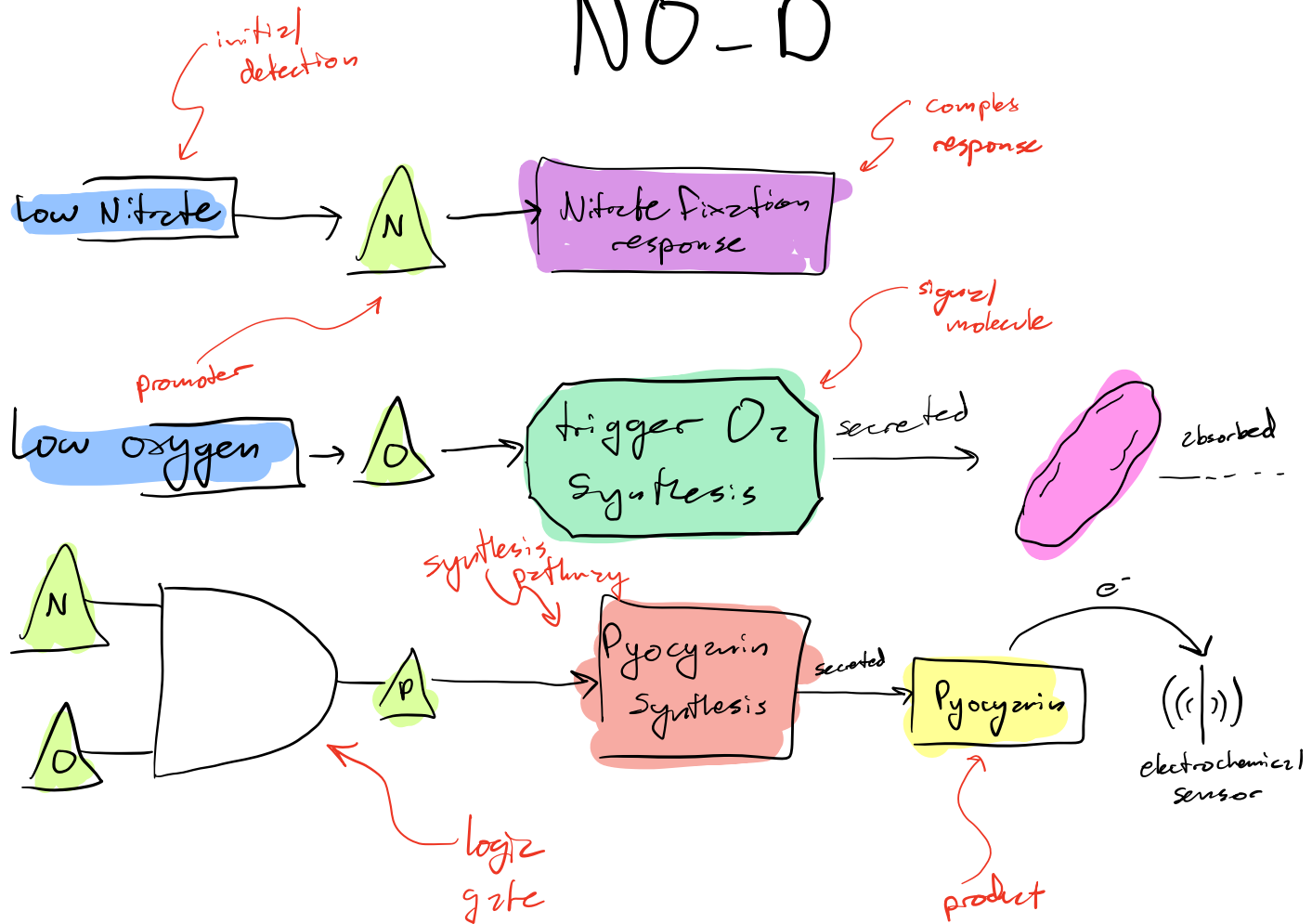
LAYER 1: Physical Environment



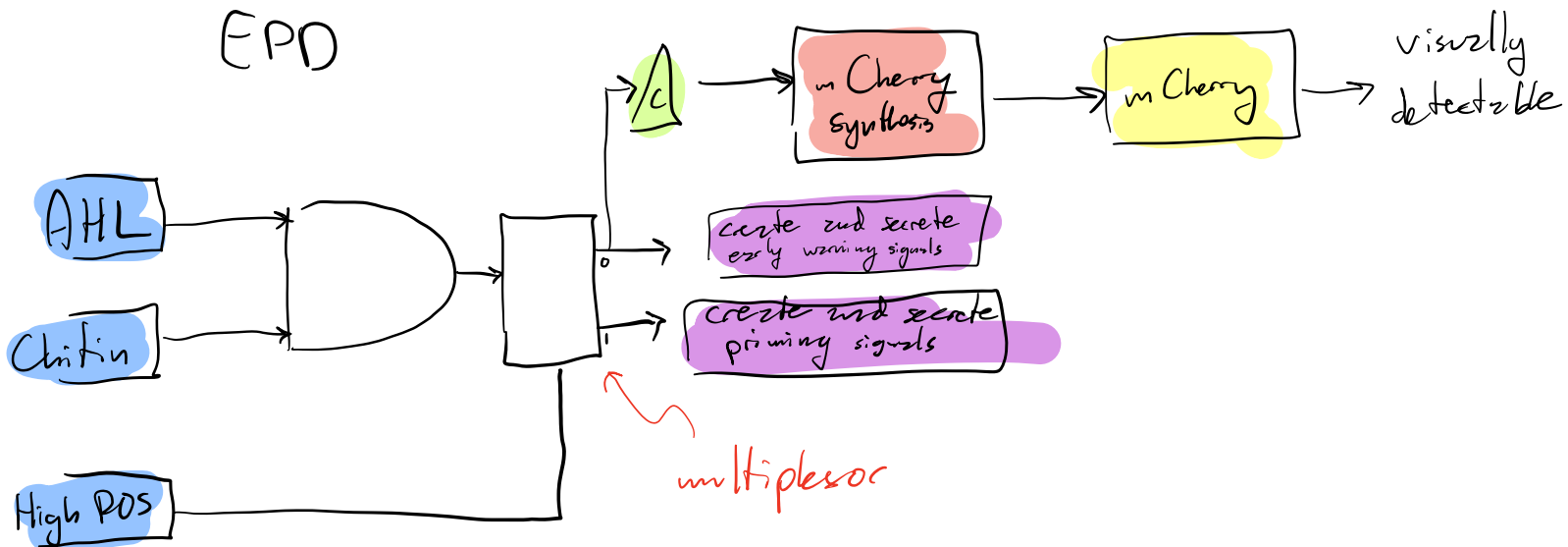
LAYER 2:

Logical Layer

NO-D



EPD



Reframing Microbes as Active Edge Processors

The concept of microbial edge computing reframes microbes not as passive biosensors, but as autonomous, programmable processors and actuators capable of building decentralized biological infrastructures. Despite its promise, MEC faces several critical hurdles. Complex genetic circuits can suffer from latency, signal noise, and stochastic failure. Microbial systems are still very fragile, as they are susceptible to mutations, environmental shifts, and cross-talk between genetic circuits. Unlike electronic circuits, there is no standardization or debugging tools for genetic circuits. Finally, the moral crumple zone remains unresolved: if a PBDU autonomously makes a harmful decision, it is unclear where responsibility should fall.

Even so, MEC offers a powerful shift in how we conceptualize biological design. While current bio-IoT frameworks treat microbes as signal reporters, MEC positions them as agents of computation and control. Advancing this paradigm could yield more robust genetic toolkits, therapeutic designs with built-in decision logic, and further biological innovations based on computer and electrical engineering principles. If scaled, MEC could enable point-of-care diagnostics in low-resource settings, fully autonomous agricultural infrastructure, and biohybrid systems capable of real-time remediation. Future integrations with endosymbiosis may allow persistent living sensors within host organisms, while future technologies like quantum communication could drastically reduce inter-microbe latency across distributed networks.

Whether or not MEC becomes a fully deployable platform, it challenges how we frame biological computation and function, while reaffirming the value of conceptual frameworks. Ideas like this push us to reconsider the assumed roles of technologies, revealing how unique properties emerge when using unconventional design approaches.

Sources

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